Evaluating Circuits (like SPICE simulator)

This Python evalSpice.py solves only DC circuits using the nodal analysis method. It takes a circuit description file as input and provides node voltages and also current through the source in a particular direction as output.

Input Format

The input file should contain a circuit description with the following elements only :

- Resistors (R)
- Voltage Sources (V)
- Current Sources (I)

the file format should be as a SPICE netlist.

Output Format

The code will return two dictionaries which are

1. Vs_dict{} --> Gives the values of voltages at the nodes.
2. I_dict{} --> Gives a single value of current flowing through voltage source in particular direction as instructed.

Solution Approach

The code utilizes the nodal analysis method to solve DC circuits. The steps involved are as follows:

1. Parsing Circuit Description: The script reads the circuit description file, extracting relevant information.
2. Node Assignment: Assigns a unique node number to each distinct node as given in the file circuit.
3. Component Identification: Identifies and categorizes components (Resistors, Voltage Sources, Current Sources) based on their labels.
4. Creating Nodal Equations: Formulates nodal equations based on Kirchhoff's Current Law (KCL) for each non-reference node.(I considered GND as zero)
5. Assembling Matrices: Constructs the matrix equation (Ax = B) for the circuit, where 'A' is the conductance matrix, 'x' is the vector of unknowns (node voltages and source currents), and 'B' is the right-hand side vector which include curent and voltage sources at relevent postions.
6. Solving Linear Equations: Utilizes NumPy's linear algebra solver to find the solution vector 'x'.
7. Updating values: Updates the node voltages and required current values accordingly into the dictionaries as asked.

Error Handling

- If the specified file does not exist, the script will raise a `FileNotFoundError`.
- If the circuit description is malformed (missing `.circuit` or `.end`), it will raise a `ValueError`.
- If there is no solution to the circuit, it will raise a `ValueError`.

Disadvantages

Does not work for the circuits which have inductors,capacitors and AC sources i.e this circuit works only for real elements.